TABLE IV

PARAMETER ESTIMATES OF THE DYNAMIC MODEL[a]

Earnings Effects $\xi_2(k')$ of Future Firm Classes								
$k'=2$	$k'=3$	$k'=4$	$k'=5$	$k'=6$	$k'=7$	$k'=8$	$k'=9$	$k'=10$
−0.005	0.004	0.005	0.022	0.002	0.015	0.009	0.016	0.023
(0.008)	(0.009)	(0.011)	(0.012)	(0.011)	(0.010)	(0.011)	(0.012)	(0.012)
Earnings Effects $\xi_3(k)$ of Past Firm Classes								
$k=2$	$k=3$	$k=4$	$k=5$	$k=6$	$k=7$	$k=8$	$k=9$	$k=10$
0.051	0.038	0.045	0.061	0.040	0.072	0.058	0.087	0.090
(0.016)	(0.015)	(0.015)	(0.016)	(0.017)	(0.015)	(0.018)	(0.016)	(0.017)

Persistence Parameters ρ			
$\rho_{1\|2}$	$\rho^m_{3\|2}$	$\rho^s_{3\|2}$	$\rho_{4\|3}$
0.227	0.246	0.681	0.651
(0.009)	(0.044)	(0.022)	(0.004)

[a]Estimates of the dynamic model, on 2001–2005. $\rho^m_{3|2}$ is the autoregressive coefficient of log-earnings for job movers between 2002 and 2004, and $\rho^s_{3|2}$ is the coefficient for job stayers. $\xi_2(k')$ and $\xi_3(k)$ denote the mean effects on log-earnings before and after a job move between firm classes k and k', respectively. $k'=1$ and $k=1$ are the omitted categories. We report parametric bootstrap standard errors in parentheses (computed using 200 replications).

6.1. *Parameter Estimates*

We estimate the dynamic model on 2001–2005, focusing on males fully-employed in the same firm in 2001–2002 and 2004–2005. In order to estimate firm classes, we use the same weighted k-means algorithm as in the static model. We then estimate the model in three steps, as explained in Section 4. We estimate the earnings persistence parameters $\rho_{4|3}$ and $\rho_{1|2}$, the wage functions and type probabilities of job movers, and those of job stayers, in turn.

In Table IV, we show estimates of several parameters of the dynamic model. The parameter $\xi_2(k')$ is the effect of firm class k' on the mean log-earnings in period 2 of a worker moving from k in period 2 to k' in period 3. It would be zero for all k' under the strict exogeneity assumption on mobility, which is imposed in our static model and many models in the literature. This effect is quantitatively small, with at most a 2% effect relative to the omitted class $k'=1$. Note that strict exogeneity of mobility also imposes that earnings realizations are independent of the subsequent decision to move. We find stronger support against this implication of exogenous mobility; see Table VII.

The parameter $\xi_3(k)$, in turn, captures the effect of firm class k on the mean log-earnings in period 3 of a worker moving between k and k'. Our static model, and many models in the literature, would also rule out the presence of a direct effect of the previous employer on current earnings. This effect appears empirically quite large in the dynamic model. It is approximately monotonic in firm class, and amounts to a 9% wage premium in the highest classes. This suggests that past firms have an impact on future earnings.

In the bottom panel of Table IV, we report the estimates of earnings persistence parameters. Persistence estimates are higher for job stayers than for job movers. Note that the autoregressive coefficient of 0.246 upon job move is significantly different from zero, which suggests that the conditional independence assumption of the static model does not hold in our data. As an important robustness check, we have estimated the persistence parameters based on covariance restrictions in first differences, in addition to the

14680262, 2019, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.3982/ECTA15722 by Tel Aviv University, Wiley Online Library on [09/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE V

VARIANCE DECOMPOSITION AND REALLOCATION EXERCISE IN THE DYNAMIC MODEL

Variance Decomposition (×100)				
$\frac{\mathrm{Var}(\alpha)}{\mathrm{Var}(y)}$	$\frac{\mathrm{Var}(\psi)}{\mathrm{Var}(y)}$	$\frac{2\,\mathrm{Cov}(\alpha,\psi)}{\mathrm{Var}(y)}$	$\frac{\mathrm{Var}(\varepsilon)}{\mathrm{Var}(y)}$	$\mathrm{Corr}(\alpha,\psi)$
60.27	4.24	13.40	22.09	41.90
(1.30)	(0.65)	(0.37)	(0.69)	(2.35)
Reallocation Exercise (×100)				
Mean	Median	10%-Quantile	90%-Quantile	Variance
0.26	0.80	2.57	−3.24	−1.05
(0.28)	(0.19)	(0.70)	(0.57)	(1.03)

[a]Estimates of the dynamic model, on 2001–2005. See the notes to Table II.

specification in levels shown in Table IV. The literature has documented differences between level estimates and first difference estimates of the dynamics of earnings in several data sets (Daly, Hryshko, and Manovskii (2016)). We find $\rho_{1|2} = 0.506$, $\rho_{4|3} = 0.451$, $\rho^m_{3|2} = 0.194$, and $\rho^s_{3|2} = 0.605$. Despite the differences in persistence estimates, variance decomposition results are similar in this case.

The estimates of the dynamic model deliver similar cross-sectional patterns for log-earnings and sorting as in the static case. In particular, the model implies approximate additivity of log-earnings in worker types and firm classes, relatively small differences across firms for all worker types except the lowest one, and strong evidence of association between worker types and firm classes (see Figure S7 in the Supplemental Material). In Table V, we report the cross-sectional variance decomposition and reallocation exercise based on the dynamic model. The variance decomposition is quite similar to the one we obtained using the static model, with some differences: the contribution of firm effects increases from 2.6% to 4.2%, and the correlation between worker and firm effects decreases from 49% to 42%. As in the static model, adding interactions between worker types and firm classes has small effects on the R^2 of the regression (i.e., 78.5% versus 77.9%). In the bottom panel of Table V, we show that the distributional effects of randomly reallocating workers across firms are also in line with the static results. The reallocation has a positive effect on mean (now insignificant) and median (significant) log-earnings, with asymmetric effects on the two tails of the distribution.

In Table VI, we show that our estimates are stable across a range of specifications, when estimating the persistence parameters in differences instead of levels (panel B), varying the numbers of firm classes and worker types (panels C and D), and estimating regression models (panel E). Using the dynamic model to reclassify firms, we obtain a larger variance contribution of firm effects and a lower correlation coefficient (panel F). This suggests that reclassifying firms does make a difference in the dynamic case, although the overall magnitudes are still quite comparable to the baseline two-step estimates of the dynamic model. Last, our framework delivers estimates of workers' mobility patterns among firms, which we report in the Supplemental Material (see Figure S8).

6.2. *Dynamic Effects*

While the dynamic and static models have similar cross-sectional implications, the richer setting we consider in this section allows us to study dynamic aspects of worker

14680262, 2019, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.3982/ECTA15722 by Tel Aviv University, Wiley Online Library on [09/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE VI

VARIANCE DECOMPOSITION (×100), DYNAMIC MODEL, ROBUSTNESS CHECKS[a]

	$\frac{\text{Var}(\alpha)}{\text{Var}(y)}$	$\frac{\text{Var}(\psi)}{\text{Var}(y)}$	$\frac{2\text{Cov}(\alpha,\psi)}{\text{Var}(y)}$	$\frac{\text{Var}(\varepsilon)}{\text{Var}(y)}$	$\text{Corr}(\alpha, \psi)$
			A. Baseline model		
	60.3	4.2	13.4	22.1	41.9
			B. Persistence parameters estimated in first differences		
	58.7	4.5	13.0	23.9	40.1
			C. Varying the number of firm classes		
$K = 3$	63.5	3.2	10.9	22.4	38.4
$K = 15$	58.7	4.9	13.8	22.6	40.8
			D. Varying the number of worker types		
$L = 3$	18.7	16.6	10.7	54.0	30.2
$L = 5$	57.8	5.5	13.6	23.2	38.0
$L = 9$	60.0	4.8	13.4	21.9	39.5
			E. Regression models		
Interactive	60.8	4.9	14.9	19.4	43.1
Linear	72.9	2.9	11.2	13.0	38.9
			F. Reclassification (starting at baseline)		
1 iteration	58.0	5.8	12.9	23.3	35.0
10 iterations	57.9	7.1	11.5	23.3	28.4

[a]Estimates of the dynamic model, on 2001–2005. See the notes to Table III.

mobility and earnings, which are interesting from both empirical and theoretical perspectives. We discuss endogenous mobility and dynamic dependence in turn.

Endogenous Mobility

We first study how the current wage affects a worker's decision to move, and which firm she moves to. In models where there is no match heterogeneity and mobility is efficient, the wage before the move should not have an effect on the propensity to move, since a worker would always move toward higher-surplus firms irrespective of her current wage. In contrast, in wage posting models with match-specific heterogeneity, a worker is more (respectively, less) likely to accept job offers when her current match is of low (resp. high) quality. In addition, endogenous mobility is ruled out in AKM fixed-effects regressions.

In Table VII, we report the estimated probability of a job move conditional on the firm class at origin, overall and by firm class at destination. In the top panel, we show those numbers in the full sample, while in the middle and bottom panels, we select workers for whom the earnings rank before the move, given firm class and worker type, is below 10 or above 90 percent, respectively. In the first column, we see that while the overall mobility rate lies between 2% and 3%, it is substantially higher for workers who had a low earnings realization in 2002. This suggests that workers are more likely to move when paid less. Such evidence of endogenous mobility is in line with estimates in Abowd, McKinney, and Schmutte (2018), and it is consistent with the predictions of wage posting models with match-specific heterogeneity. In contrast, our estimates in the bottom panel suggest that high earnings realizations do not strongly affect mobility. Last, results by firm class at

14680262, 2019, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.3982/ECTA15722 by Tel Aviv University, Wiley Online Library on [09/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE VII

TRANSITION PROBABILITIES ($\times 100$), BY CONDITIONAL DECILE OF PREVIOUS EARNINGS[a]

	All movers	$k' = 1$–3	$k' = 4$–7	$k' = 8$–10
		All		
$k = 1$–3	2.20	0.84	1.06	0.30
	(0.03)	(0.02)	(0.03)	(0.02)
$k = 4$–7	1.86	0.39	1.03	0.44
	(0.03)	(0.02)	(0.02)	(0.01)
$k = 8$–10	2.90	0.47	1.00	1.43
	(0.06)	(0.06)	(0.03)	(0.02)
		First Conditional Decile of Earnings		
$k = 1$–3	3.41	1.53	1.52	0.37
	(0.21)	(0.09)	(0.12)	(0.05)
$k = 4$–7	3.20	0.77	1.74	0.69
	(0.17)	(0.09)	(0.09)	(0.05)
$k = 8$–10	4.92	0.93	1.74	2.25
	(0.31)	(0.15)	(0.14)	(0.13)
		Tenth Conditional Decile of Earnings		
$k = 1$–3	2.76	0.95	1.42	0.40
	(0.20)	(0.08)	(0.13)	(0.06)
$k = 4$–7	1.82	0.35	1.05	0.42
	(0.10)	(0.04)	(0.06)	(0.03)
$k = 8$–10	2.03	0.29	0.71	1.03
	(0.14)	(0.06)	(0.06)	(0.08)

[a]We report the probability of changing job, overall and by destination firm class k', for each origin firm class k. In the top panel, we show results for all workers. In the middle and bottom panels, we show results for workers in the first and tenth deciles of log-earnings Y_{i2}, respectively, conditional on worker type α_i and current firm class $k_{i2} = k$. The estimates are obtained using the dynamic model, with 10,000,000 simulations. We report parametric bootstrap standard errors in parentheses (computed using 200 replications).

destination do not seem to vary much with earnings realizations, which is in line with the small estimates of $\xi_2(k')$ that we found.

State Dependence and Network Effects

We next use our dynamic estimates to study how a worker's wage is affected by the firms she works for over time. Specifically, we focus on how much of the wage dispersion among similar workers is explained by the workers' previous employers. There are two different reasons why the previous employer may matter. First, working in a high-wage firm, say, may make a worker more likely to move to another high-wage firm. We refer to this effect, which is driven by mobility patterns across the network of firms, as the *network effect*. Second, the past firm may have a direct effect on the worker's wage after a job move. We call this the *state dependence effect*.

Network and state dependence effects are present in many models of earnings and mobility. Consider, as an example, sequential bargaining models where firms are characterized by their productivities (e.g., Postel-Vinay and Robin (2002), Lise, Meghir, and Robin (2016), Bagger and Lentz (2014)). Due to the on-the-job search process, the distribution of productivities of future employers depends on the productivity of the current employer (network effect). Moreover, due to the offer and counteroffer mechanism, a worker coming from a more productive firm is able to extract a higher share of the sur-

14680262, 2019, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.3982/ECTA15722 by Tel Aviv University, Wiley Online Library on [09/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License